

Deep Learning and Causal AI for Estimation

Lecture 0: Introduction

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Google Trends: “AI”

Interest over time ^①

Worldwide · 2004 - present



Distinct Sets of Questions

AI Impact on Economy

Labor market effects

Productivity implications

Organization of work

Distinct Sets of Questions

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AI Impact on Economics Profession

Coding agents and the research pipeline

Automation of paper production/referee reports

Which skills to develop in classroom

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Which skills to develop in classroom

AI Impact on Measurement and Inference

This course

Examples

1. [Baker et al., 2016]
economic policy uncertainty; data: newspaper text.
2. [Gentzkow et al., 2019b]
polarization; data: Congressional Records.
3. [Bandiera et al., 2020]
CEO behavior; data: time use surveys.
4. [Adukia et al., 2023]
cultural representation; data: children's book photos.
5. [Gorodnichenko et al., 2023]
Fed chairperson tone; data: FOMC press conference audio feed.

GPT as a Measurement Tool

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WORKING PAPER 34834

DOI 10.3386/w34834

ISSUE DATE February 2026

We present the GABRIEL software package, which uses GPT to quantify attributes in qualitative data (e.g. how "pro innovation" a speech is). GPT is evaluated on classification and attribute rating performance against 1000+ human annotated tasks across a range of topics and data. We find that GPT as a measurement tool is accurate across domains and generally indistinguishable from human evaluators. Our evidence indicates that labeling results do not depend on the exact

20:30



Search



Suggested



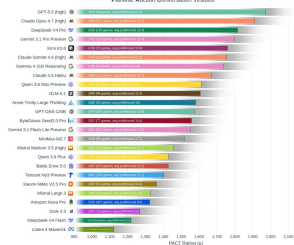
Ethan Mollick · 2nd
Associate Professor at The Wharton Sc...
3d ·

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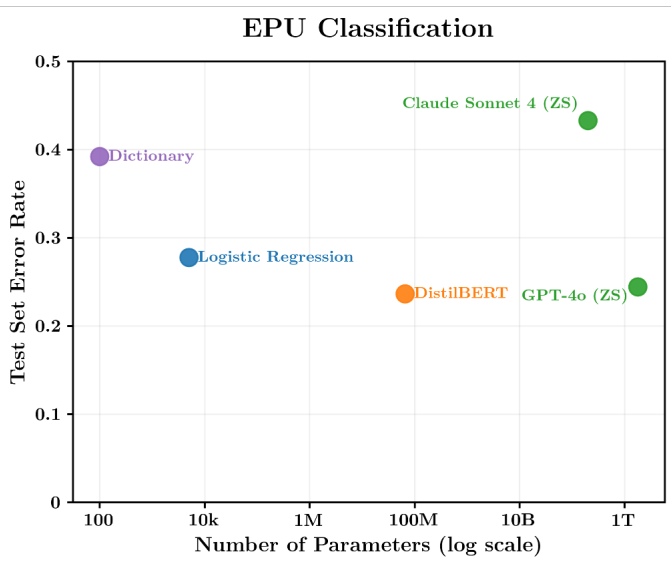
One of the most important properties of LLMs that we take for granted is that newer, bigger models are just generally better at everything. The AI Labs are pouring effort into economically valuable fields like coding, but bigger models are also just happen to be better at negotiation, alignment, strategy, poetry, etc.

It is one reason why I think the push for smaller, local models is more complicated than people think. If you want good answers, especially good answers to unexpected problems, frontier models will generally outperform smaller models on a wide range of tasks.

PACT: Benchmarking LLM negotiation skill in multi-round buyer-seller bargaining
Pairwise Auction Conversation Testbed



The Reality



Lecture Plan

Lecture 1: Neural network overview

Lecture 2: Foundation LLM/AI models.

Lecture 3: Adapting foundation models for economic measurement.

Lecture 4: AI/ML-generated estimates in regression models.

All material on

https://github.com/sekhansen/fra_iese_course_2026.

Background Reading

[Gentzkow et al., 2019a]

[Ash and Hansen, 2023]

[Ash et al., 2025]

[Ludwig et al., 2025]

[Jurafsky and Martin, 2025] <https://stanford.io/4qkv65V>

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